Intermediate Computer Science 30-Day Crash

Two hours a day · board, drill, mark scheme · all six units

DAYS 1–12 SYSTEMS | DAYS 13–18 ALGORITHMS | DAYS 19–24 PYTHON | DAYS 25–28 DATA & IMPACT | DAYS 29–30 MOCK

How a coaching day works. Recall yesterday with the book shut. Copy the board notes. Work the example in pencil before you read the steps. Sit the drill in silence, then mark in a second colour. The green lines are the mark scheme, not a script to memorise word for word.

Missed a day? Do that day’s drill only, then rejoin the timetable. Do not stack three new topics in one evening.

Pair with the Teach Yourself lecture notes. This booklet is the timed revision, not a second textbook.

The thirty days on one sheet

Each row is a 120-minute class. Checkpoint days are closed-book. Day 29 is a full mock. Day 30 is the marking conference.

Class XI crash timetable

Day	Unit	Class target	Score
01	Systems	Analog, digital, bits	/5
02	Systems	AND, OR, NOT tables	/5
03	Systems	Seven gates from memory	/5
04	Systems	Precedence and NAND/NOR	/5
05	Systems	SOP and POS	/5
06	Systems	2-var and 3-var K-maps	/5
07	Systems	LogiSim on paper and screen	/5
08	Systems	Six SDLC phases	/5
09	Systems	Al-Noor: Waterfall vs Agile	/5
10	Systems	OSI seven layers and PDUs	/5
11	Systems	TCP/IP and encapsulation	/5
12	Systems	Unit 1 checkpoint	/10
13	Algorithms	Decompose, pattern, abstract	/5
14	Algorithms	Bubble sort, full trace	/5
15	Algorithms	Selection sort	/5
16	Algorithms	Linear and binary search	/5
17	Algorithms	Correct, efficient, clear	/5
18	Algorithms	Unit 2 checkpoint	/10
19	Python	Types, input, print	/5
20	Python	Operators, including // and %	/5
21	Python	if / elif / else	/5
22	Python	for, while, break, continue	/5
23	Python	Libraries and three bug families	/5
24	Python	Unit 3 checkpoint, on paper	/10
25	Data	Keys, 1:M, M:N	/5
26	Data	Library ER and Access objects	/5
27	Impacts	IoT loop, sources, assistive tech	/5

28	Literacy	Research question to artefact	/5
29	Mock	90-minute mixed paper	/50
30	Mock	Mark scheme and 48-hour card	—

PASS MARK A checkpoint under 7/10 means that unit is not safe. Repeat its drills on the next evening before you touch new work. The mock target is 35/50 with no blank long question.

DAY 01

Analog, digital, and the bit

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Separate discrete from continuous
- Sketch sine and square waves
- State what 0 and 1 mean in a digital system

TRAP THAT COSTS THE MARK

Calling every number on a screen ‘analog’. A digital display can show 37.4 and still be a digital system.

Board notes

A **discrete** quantity jumps between countable values: students in a room, books on a shelf. A **continuous** quantity can sit anywhere in a range: temperature, height, speed. Stairs are discrete. A ramp is continuous. Digital machines chose the stairs.

A **signal** carries information. An analog signal changes smoothly (voice, guitar, old telephone voltage) and is easily bent by noise. A digital signal has only HIGH and LOW, so a slightly dented pulse is still read as a bit.

$$0 = \text{OFF / LOW / FALSE}, \qquad 1 = \text{ON / HIGH / TRUE}$$

	Analog	Digital
Values	Any value in a range	Two levels
Wave	Smooth	Square
Noise	Corrupts the value	Usually ignored
Examples	Voice, vinyl, mercury thermometer	PC, phone, USB, HDMI

Worked example — do not skip the steps

Board. A clinic stores body temperature. The mercury thermometer is analog. The digital thermometer stores 37.4 after rounding to one decimal — still a number on a screen, but the electronics inside use bits. Ask: which representation survives a noisy cable? The bit stream. Which one can show a value between two marks without rounding? The analog instrument.

Write one sentence a marker can award: “Digital systems use two states so copies stay exact and noise rarely flips a solid HIGH into a solid LOW.”

Silent drill — cover the green model lines

1. Define discrete and continuous with one example each.

Model. Discrete: separate countable values (fingers on a hand). Continuous: any value in a range (speed of a car).

2. Why did engineers accept only two voltages?

Model. Two states are reliable against noise, exact when copied, simple to design, and easy to program.

3. Sketch labels, not art: which wave is HDMI?

Model. A square wave. HDMI is a digital signal, only HIGH and LOW.

4. Is ‘number of cars in a lot’ analog or discrete?

Model. Discrete. You cannot have 2.3 cars. The count jumps by whole cars.

5. A song on vinyl versus the same song as a file. One reliability difference.

Model. Vinyl (analog) picks up hiss; the file (digital) copies without that quality loss.

TONIGHT Redraw both waves from memory and label three real devices under each.

DAY
02

Boolean algebra: AND, OR, NOT

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Build 4-row truth tables
- Use Boolean + as OR
- State precedence

TRAP THAT COSTS THE MARK

Treating $+$ as ordinary addition. In this unit $1+1$ is 1.

Ten-minute recall — yesterday, closed book

Ask	Accept
What does bit 0 mean?	OFF, LOW, or FALSE
Square wave or sine wave for USB?	Square / digital

Board notes

George Boole's algebra has two constants. Variables such as A and B hold only those constants. There are three basic operations.

Op	Symbol	Rule
AND	$A \cdot B$	1 only when every input is 1
OR	$A + B$	1 when any input is 1
NOT	$\overline{A}$ or A'	Flips the single input

$$1 + 1 = 1 \quad \text{(OR, not arithmetic)}$$

Precedence, tightest first: brackets, then NOT, then AND, then OR. So $A + B \cdot C$ means $A + (B \cdot C)$.

Worked example — do not skip the steps

Board: $Y = \overline{A} + B$. Four rows, because $2^2 = 4$.

A	B	$\overline{A}$	Y
0	0	1	1
0	1	1	1
1	0	0	0
1	1	0	1

The only 0 is row $A = 1, B = 0$. Say the English: "Y is 0 only when A is true and B is false."

Silent drill — cover the green model lines

1. How many rows does a truth table with 3 inputs have?

Model. 8, because $2^3 = 8$. List inputs in binary counting order so no row is skipped.

2. Compute $1 + 1$ and $1 \cdot 0$ in Boolean algebra.

Model. OR: $1 + 1 = 1$. AND: $1 \cdot 0 = 0$.

3. Door with two locks. Which operation?

Model. AND. Both locks must be open. $Y = A \cdot B$.

4. Room with two doors. Which operation?

Model. OR. Either door is enough. $Y = A + B$.

5. Evaluate $A + B \cdot C$ when $A = 0, B = 1, C = 0$.

Model. AND first: $B \cdot C = 0$. Then OR: $0 + 0 = 0$.

TONIGHT Write the AND and OR tables from a blank page. Time yourself: four minutes.

DAY
03

Seven gates from a blank page

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Draw AND, OR, NOT, NAND, NOR, XOR, XNOR
- Match each gate to an English rule
- Name the two universal gates

TRAP THAT COSTS THE MARK

Swapping XOR and OR. OR is 1 when any input is 1, including both. XOR is 0 when both are 1.

Ten-minute recall — yesterday, closed book

Ask	Accept
Boolean 1+1?	1
Rows for 2 inputs?	4

Board notes

A gate is the circuit that performs one Boolean operation. Learn shape, expression, and the sentence “output is 1 when …” together. A bubble on the output means invert.

Gate	Expression	Output is 1 when
AND	$Y = A \cdot B$	All inputs are 1
OR	$Y = A + B$	Any input is 1
NOT	$Y = \overline{A}$	Input is 0
NAND	$Y = \overline{A \cdot B}$	Any input is 0
NOR	$Y = \overline{A + B}$	All inputs are 0
XOR	$Y = A \oplus B$	Inputs differ
XNOR	$Y = \overline{A \oplus B}$	Inputs are the same

NAND and NOR are **universal**: any Boolean function can be built from only NAND, or only NOR.

Worked example — do not skip the steps

Board stories the marker likes.

- Car starts: key AND start button.
- Alarm: front door OR back door.
- Two-way staircase light: XOR. The lamp is on when the switches disagree.
- Equality check: XNOR. Output 1 when both bits match.

Draw NAND as an AND shape plus a bubble. Draw XOR as an OR shape plus an extra curved line on the input side. Practise until the seven outlines take under two minutes.

Silent drill — cover the green model lines

1. XOR of 1 and 1?

Model. 0. XOR is 1 only when the inputs differ.

2. NOR of 0 and 0?

Model. 1. NOR is 1 only when every input is 0.

3. Which two gates are universal?

Model. NAND and NOR.

4. XNOR expression in words.

Model. NOT XOR, or 1 when inputs are equal: $AB + \overline{A}\,\overline{B}$.

5. A bubble on the output of AND produces which gate?

Model. NAND.

> **TONIGHT** Draw all seven gates, unlabelled, then label them tomorrow morning without looking.

DAY
04

Expressions, precedence, universal gates

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Evaluate a mixed expression row by row
- Build NOT, AND, and OR from NAND
- Draw $Y = (A + B) \cdot \overline{C}$

TRAP THAT COSTS THE MARK

Forgetting the bubble when you copy a NAND into a diagram, which silently turns it into AND.

Ten-minute recall — yesterday, closed book

Ask	Accept
Universal gates?	NAND and NOR
XOR of 1 and 0?	1

Board notes

An expression is variables joined by operations. Add a column to the truth table for every sub-expression, left to right after applying precedence.

$$Y = (A + B) \cdot \overline{C}$$

Order inside that formula: the bracketed OR, the NOT on C, then the AND. Three inputs means eight rows.

From NAND alone. NOT: $\overline{A} = \overline{A \cdot A}$ (tie both NAND inputs to A). AND: NAND followed by NOT. OR needs De Morgan: $A + B = \overline{\overline{A} \cdot \overline{B}}$, which is a NAND of two NOTs. You do not need the algebra name in the answer if you show the gates.

Worked example — do not skip the steps

Board row. $A = 1, B = 0, C = 1$ in $Y = (A + B) \cdot \overline{C}$.

1. $A + B = 1 + 0 = 1$.
2. $\overline{C} = \overline{1} = 0$.
3. $Y = 1 \cdot 0 = 0$.

Second row to speak aloud: $A = 0, B = 0, C = 0$ gives $0 \cdot 1 = 0$. Third: $A = 1, B = 1, C = 0$ gives $1 \cdot 1 = 1$.

Silent drill — cover the green model lines

1. Evaluate $A \cdot B + \overline{C}$ at $A = 1, B = 1, C = 1$.

Model. AND first: $1 \cdot 1 = 1$. NOT C is 0. OR: $1 + 0 = 1$.

2. How do you make a NOT from one NAND?

Model. Join both inputs of the NAND to the same variable. Output is $\overline{A}$.

3. Why eight rows for $Y = (A + B) \cdot \overline{C}$?

Model. Three input variables, and $2^3 = 8$.

4. Which operation is done first in $A + B \cdot C$ if there are no brackets?

Model. AND. The expression means $A + (B \cdot C)$.

5. Name the gate for ‘inputs different’.

Model. XOR.

TONIGHT Pick one 3-variable expression and finish all eight rows. Do not stop at the interesting row.

DAY 05

Minterms, maxterms, SOP and POS

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Write a minterm for a row
- Build SOP from the 1-rows
- Build POS from the 0-rows

TRAP THAT COSTS THE MARK

Numbering minterms out of binary order. m_2 is 10, not 01.

Ten-minute recall — yesterday, closed book

Ask	Accept
Precedence winner between AND and OR?	AND
Rows for 3 inputs?	8

Board notes

A **minterm** is an AND that contains every variable. If the variable is 0 in that row, write it complemented. You OR the minterms of every row where $Y = 1$. That is the **sum of products (SOP)**.

A **maxterm** is an OR that contains every variable, complemented when the variable is 1 in that row. You AND the maxterms of every row where $Y = 0$. That is the **product of sums (POS)**.

$$m_0 = \overline{A}\,\overline{B},\ m_1 = \overline{A}B,\ m_2 = A\overline{B},\ m_3 = AB$$

The textbook prefers SOP when you draw the circuit: AND gates into one OR gate.

Worked example — do not skip the steps

Board. $Y = 1$ only on rows $(0,1)$ and $(1,1)$.

SOP: $Y = \overline{A}B + AB$. Factor: $Y = B(\overline{A} + A) = B$. Check: both 1-rows have $B = 1$, both 0-rows have $B = 0$. The simpler answer is $Y = B$.

POS from the 0-rows $(0,0)$ and $(1,0)$: $Y = (A + B)(\overline{A} + B)$, which is also B. Either form is full marks if the rows are correct; the simplified form is the one you should draw.

Silent drill — cover the green model lines

1. Write m_2 for variables A, B.

Model. $A\overline{B}$. Row 2 is $A = 1, B = 0$.

2. SOP uses which rows?

Model. The rows where output $Y = 1$.

3. POS uses which rows?

Model. The rows where output $Y = 0$.

4. $Y = \overline{A}B + AB$. Simplify.

Model. $Y = B$.

5. Why does the course draw SOP as gates?

Model. Each product is an AND, and the sum of those products is one OR.

TONIGHT From a blank 2-variable table with 1s only at m_0 and m_2, write SOP and simplify to $\overline{B}$.

DAY
06

Karnaugh maps

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Fill a map in Gray code
- Group power-of-two cells
- Read a group back as a term

TRAP THAT COSTS THE MARK

Grouping three cells. Three is not a power of two; split into a pair plus a single, or find a pair and another pair.

Ten-minute recall — yesterday, closed book

Ask	Accept
SOP comes from which output rows?	Y=1 rows
What is a minterm?	AND of every variable, complemented where the row has 0

Board notes

A K-map is a truth table redrawn so neighbours differ by one bit (Gray code). Maurice Karnaugh, 1953. You circle groups of size $1, 2, 4, 8, \ldots$ — powers of two. Bigger groups make shorter expressions. Overlap is allowed. Edges wrap around, including corners on a 3-variable map.

Column order for three variables is $BC = 00, 01, 11, 10$. Never $00, 01, 10, 11$: that last pair is not adjacent in Gray code.

Inside a circle, drop any variable that changes, and keep any variable that stays constant (complemented if it stays 0).

Worked example — do not skip the steps

Board, two variables. Cells: $m_0 = 0, m_1 = 1, m_2 = 1, m_3 = 1$. That is $Y = \overline{A}B + A\overline{B} + AB$.

Group the bottom row ($A = 1$, both columns) as A. Group the right column ($B = 1$, both rows) as B. Overlap on AB is legal. Result: $Y = A + B$. Check against the OR table: the only 0 is $A = B = 0$.

Three-variable collapse the book likes. If every 1-cell contains C and the pairs cover all of A and B, the whole group reads $Y = C$.

Silent drill — cover the green model lines

1. Legal group sizes?

Model. Powers of two: 1, 2, 4, 8, …

2. Gray-code order of two bits?

Model. 00, 01, 11, 10.

3. May groups overlap?

Model. Yes. Overlap often gives the simplest cover.

4. A 2-cell group in which A changes and B stays 1. The term?

Model. B.

5. Simplify $Y = \overline{A}B + AB$ on a map or by factoring.

Model. $Y = B$.

> **TONIGHT** Draw a 2-variable map, place three 1s, circle, and write the OR expression without looking at today’s board.

DAY 07

LogiSim, including a paper lab

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Place pins and a gate
- Wire only horizontally and vertically
- Verify $Y = A \cdot B + C$ against eight rows

TRAP THAT COSTS THE MARK

Trusting the simulator when your table disagrees. The table is the specification; a crossed wire is a bug.

Ten-minute recall — yesterday, closed book

Ask	Accept
Gray code pair after 01?	11, then 10
Group size 3 allowed?	No

Board notes

LogiSim Evolution is the course simulator. It needs Java 21 or newer. Circuits save as `.circ`. Input pins are squares you toggle. Output pins are circles. Bright green means 1; dark green means 0. Wires must be horizontal or vertical.

1. File, New, Save As `and_gate.circ`.
2. Place AND. Place input pins A and B on the left, output Y on the right.
3. Wire pin to gate. Poke each of the four input pairs. Y is 1 only for $(1, 1)$.
4. Extension: AND of A and B, then OR with C. Tick all eight rows.

No lab PC today? Draw the same circuit on paper and fill the eight-row table. The logic mark does not require the software screenshot if the table matches.

Worked example — do not skip the steps

Board checklist for $Y = A \cdot B + C$.

A	B	C	$A \cdot B$	Y
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	1

Y is 0 only when $C = 0$ and the AND is 0. Say that sentence before you poke the circuit.

Silent drill — cover the green model lines

1. File extension of a LogiSim circuit?

Model. `.circ`.

2. Which pin do you toggle, square or circle?

Model. Square input pin. The circle is the output.

3. May a wire run diagonally?

Model. No. Horizontal or vertical only.

4. Colour of logic 1 on the wire?

Model. Bright green. Dark green is 0.

5. For $Y = A \cdot B + C$, what is Y when $A = B = C = 0$?

Model. 0.

> **TONIGHT** If a machine is available, build the AND and photograph the $(1, 1)$ case. If not, reproduce the eight-row table from memory.

DAY 08

The six SDLC phases

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Name each phase and its deliverable
- Contrast black-box and white-box tests
- Name four maintenance types

TRAP THAT COSTS THE MARK

Putting coding before an SRS because ‘we already know what to build’.

Ten-minute recall — yesterday, closed book

Ask	Accept
LogiSim wire rule?	Horizontal or vertical only
Bright green means?	Logic 1

Board notes

The software development life cycle is the agreed sequence so a team does not pour the concrete and then ask how many rooms. Six phases:

Phase	Deliverable
Requirement analysis	SRS — software requirements specification
System design	HLD/LLD, database, wireframes
Implementation	Source code, usually in Git
Testing	Unit, integration, system, UAT; defect log
Deployment	Direct, phased, pilot, or parallel run
Maintenance	Corrective, adaptive, perfective, preventive

Black box: you see inputs and outputs, not the code. **White box:** you test branches and loops inside the code.

Worked example — do not skip the steps

Board: a fees report required by a board circular, deadline fixed, rules known. Phase 1 writes the SRS and the school signs it. Design draws tables and screens. Code follows the design. Testing includes UAT with the accounts clerk (black box) and branch tests on the calculation (white box). Deployment might be parallel: old register and new system for two weeks. Maintenance later is adaptive when the circular changes.

Silent drill — cover the green model lines

1. Which document freezes requirements?

Model. The SRS.

2. UAT sits in which phase?

Model. Testing. Users accept the system against the requirements.

3. Corrective maintenance fixes what?

Model. Defects found after release.

4. Adaptive maintenance responds to what?

Model. A changed environment or rule, such as a new fee policy.

5. Pilot deployment means what?

Model. Release to a small group first, then widen.

TONIGHT Write the six phase names in order on a card you will keep until Day 12.

DAY 09

Waterfall and Agile at Al-Noor

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Choose a model from the project facts
- Describe a sprint
- Quote one difference that markers want

TRAP THAT COSTS THE MARK

Saying ‘Agile means no documentation’. It means light documentation and working software as the proof.

Ten-minute recall — yesterday, closed book

Ask	Accept
SRS belongs to which phase?	Requirement analysis
Black box hides what?	The source code

Board notes

Waterfall finishes a phase before the next begins. Documents are heavy. Change is expensive. Use it when requirements are stable, the technology is known, and the project is small enough to plan.

Agile / Scrum ships a usable slice every sprint of about 2 to 4 weeks. The product backlog is a living list of user stories (“As a student, I want …”). A review shows working software. A retrospective improves the team.

	Library LMS	Student mobile app
Model	Waterfall	Agile, Scrum
Stack	Python, Tkinter, SQLite	Flutter
Why	Rules known, deadline fixed	Needs unclear, many stakeholders
First working build	About six months	About two weeks
Change	SRS frozen	Next sprint

Worked example — do not skip the steps

Board decision. “The school wants a bus tracker, but students, parents, and the principal disagree, and the phone OS will update twice this term.” Agile. Say the reason in one line: requirements are not knowable on day one, and a sprint review is how the school will discover them.

Opposite board: “The library already has a signed list of fields and a six-month handover.” Waterfall. The LMS case froze STUDENTS, BOOKS, and LOANS, with 14-day loans.

Silent drill — cover the green model lines

1. Name the Waterfall case and the Agile case.

Model. Al-Noor library LMS is Waterfall. Al-Noor student mobile app is Agile.

2. What is a user story?

Model. A backlog item in the user’s words: As a <role>, I want <goal>.

3. Length of one sprint in the mobile case?

Model. Two weeks. Five sprints, about ten weeks.

4. When is change cheap?

Model. In Agile, by placing it in a later sprint. In Waterfall it is expensive after the SRS is signed.

5. Testing in Agile versus Waterfall?

Model. Agile tests inside every sprint. Waterfall has a dedicated later test phase.

> **TONIGHT** Write a six-line comparison table from memory: requirements, client contact, first build, change, testing, documents.

DAY 10

OSI: seven layers and the PDU chant

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- List layers from physical up
- Attach one job and one protocol to each
- Name the data unit at each layer

TRAP THAT COSTS THE MARK

Putting IP at the data-link layer. IP is layer 3. MAC is layer 2.

Ten-minute recall — yesterday, closed book

Ask	Accept
Agile case study name?	Al-Noor student mobile app
Waterfall first working build, roughly?	Six months

Board notes

Mnemonic, bottom to top: **Please Do Not Throw Sausage Pizza Away** — Physical, Data link, Network, Transport, Session, Presentation, Application.

#	Layer	Job	PDU
7	Application	HTTP, DNS, SMTP, FTP	Data
6	Presentation	Format, encrypt, compress	Data
5	Session	Open, keep, close dialogs	Data
4	Transport	TCP or UDP, ports	Segment
3	Network	IP addresses, routing	Packet
2	Data link	MAC, frames, CRC	Frame
1	Physical	Bits on cable or radio	Bits

Data → Segment → Packet → Frame → Bits

Worked example — do not skip the steps

Board: a web page request. The browser is application (HTTP, port 443 if HTTPS). Presentation handles TLS. Transport chooses TCP because a missing byte matters. Network puts the server's IP on the packet. Data link puts the next hop's MAC on the frame. Physical sends bits. On the way down we **encapsulate** (add headers). On the way up we **decapsulate** (peel them).

TCP is reliable and ordered. UDP is faster and does not retry. Video calls often prefer UDP because a late packet is useless.

Silent drill — cover the green model lines

1. Which layer adds port numbers?

Model. Transport.

2. Which layer adds the IP address?

Model. Network.

3. Which layer adds the MAC address?

Model. Data link.

4. PDU at layer 3?

Model. Packet.

5. Why might a call use UDP?

Model. Retransmission would arrive too late; UDP keeps the stream timely.

> **TONIGHT** Recite the mnemonic and the PDU chant before sleep. Write ports 80 and 443 beside TCP.

DAY 11

TCP/IP and a packet’s journey

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Map four TCP/IP layers onto OSI
- Contrast the teaching model with the Internet model
- Walk encapsulation in five steps

TRAP THAT COSTS THE MARK

Saying OSI ‘is the Internet’. The Internet runs TCP/IP.

Ten-minute recall — yesterday, closed book

Ask	Accept
Transport PDU name?	Segment
Layer of a MAC address?	Data link

Board notes

TCP/IP is the model that actually runs the Internet: four layers, from 1970s US Department of Defense work. OSI (ISO, 1984) is the seven-layer teaching reference.

TCP/IP	OSI layers folded in	Examples
Application	7 + 6 + 5	HTTP, DNS, FTP, SMTP
Transport	4	TCP, UDP
Internet	3	IPv4, IPv6, ICMP
Network access	2 + 1	Ethernet, Wi-Fi, cables

Say this in the exam: OSI explains. TCP/IP implements.

Worked example — do not skip the steps

Board journey, college PC to a result website.

1. Application builds the HTTP request.
2. Transport wraps it as a TCP segment, destination port 443.
3. Internet wraps that as an IP packet with the server address.
4. Network access wraps a frame with the gateway MAC and sends bits.
5. Each router looks at the IP, not the web page. The page is unwrapped only at the far host.

Silent drill — cover the green model lines

1. How many layers in TCP/IP?

Model. Four.

2. Which TCP/IP layer holds Ethernet?

Model. Network access.

3. Which OSI layers does the TCP/IP application layer cover?

Model. Application, presentation, and session (7, 6, and 5).

4. Encapsulation or decapsulation on the sending host?

Model. Encapsulation: headers are added on the way down.

5. One line: OSI versus TCP/IP.

Model. OSI is a 7-layer reference for teaching. TCP/IP is the 4-layer model the Internet runs.

TONIGHT Draw four boxes, label them, and write one protocol in each. Sixty seconds.

DAY 12

Unit 1 checkpoint

Unit 1 · Computer Systems · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Score at least 7 out of 10
- Write full sentences, not single words
- Fix any gate or OSI gap tonight

TRAP THAT COSTS THE MARK

Moving on with a 4/10 because the timetable says Day 13. The timetable loses to the score.

Ten-minute recall — yesterday, closed book

Ask	Accept
TCP/IP layer count?	4
OSI layer count?	7

Board notes

Closed book. Ten short items. One mark each. Write the mark scheme only after you have answered. A score under 7 means Days 1–11 are not safe: redo the drills you missed before Day 13.

Worked example — do not skip the steps

No new example today. The checkpoint is the example. Sit it in 35 minutes, mark it in 15, and spend the rest of the class repairing the lowest topic.

Silent drill — cover the green model lines

1. State 0 and 1 in words.

Model. 0 is OFF/LOW/FALSE. 1 is ON/HIGH/TRUE.

2. Boolean value of $1 + 1$ and of $1 \cdot 0$.

Model. 1 and 0.

3. Output rule for NAND.

Model. 1 when any input is 0. Expression $\overline{A \cdot B}$.

4. Why is NOR universal? One sentence.

Model. Any Boolean function can be built using only NOR gates.

5. SOP is built from which rows?

Model. Rows where $Y = 1$, as a sum of minterms.

6. Gray-code order of 00, 01, 10, 11 corrected.

Model. 00, 01, 11, 10.

7. Name the six SDLC phases in order.

Model. Analysis, design, implementation, testing, deployment, maintenance.

8. Al-Noor library: which model, and why in one line?

Model. Waterfall, because requirements were known and then frozen.

9. PDU chant from application down to the cable.

Model. Data, segment, packet, frame, bits.

10. Which TCP/IP layer corresponds to OSI layers 2 and 1?

Model. Network access.

> **TONIGHT** List the three questions you lost. Redo only those board examples. Do not reread the whole unit.

DAY 13

Decompose, notice the pattern, abstract

Unit 2 · Computational Thinking · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Split a problem until each piece is testable
- Spot a repeating rule
- Drop details that do not change the result

TRAP THAT COSTS THE MARK

Starting with Python keywords before the steps exist.

Ten-minute recall — yesterday, closed book

Ask	Accept
Checkpoint pass mark?	7/10
PDU under the packet?	Frame, then bits

Board notes

Computational thinking is the habit before the language. Four moves: **decompose**, **recognise the pattern**, **abstract**, then write the **algorithm**. An algorithm is a language-free sequence of steps. Pseudocode is that sequence written with IF, FOR, and WHILE so it can become Python later.

Decomposition: a timetable clash is not one blob. It is “read the grids”, “find the shared period”, “propose a swap”, “check the room”. Pattern: row i of a star triangle prints i stars, so one loop replaces ten drawings. Abstraction: making tea keeps boil, add tea, steep, pour — and drops the colour of the cup.

Worked example — do not skip the steps

Board: attendance app. Decompose into login, class list, mark present or absent, save, report. Abstract away the logo colour. The pattern is “for each student, record one status”. Pseudocode:

```
FOR each student in class
    READ status
    IF status is absent THEN count = count + 1
WRITE count
```

Silent drill — cover the green model lines

1. Algorithm or pseudocode: which one may use IF and WHILE?

Model. Pseudocode. An algorithm can be plain numbered steps and is language-free.

2. One detail you abstract away in a library search.

Model. Spine colour. Keep title, author, availability.

3. Row i prints how many stars?

Model. i stars. That single rule is the pattern.

4. Why decompose?

Model. Small pieces can be tested and explained. A blob cannot.

5. Who is an algorithm for?

Model. Anyone who must follow the logic, including a non-programmer. Pseudocode aims at the person about to code.

> **TONIGHT** Take ‘prepare a practical file’ and split it into six steps a classmate could follow without you.

DAY 14

Bubble sort, every pass written out

Unit 2 · Computational Thinking · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Perform neighbour swaps
- Show the settled suffix
- State $O(n^2)$ and stability

TRAP THAT COSTS THE MARK

One pass and a claim that the list is finished. Markers want the settled value named.

Ten-minute recall — yesterday, closed book

Ask	Accept
Four thinking moves, in order?	Decompose, pattern, abstract, algorithm
Pseudocode keywords?	IF, FOR, WHILE

Board notes

Bubble sort walks left to right, compares neighbours, and swaps if they are out of order. The largest unsorted value settles at the end of the pass. The next pass ignores that settled suffix.

$$\textbf{for } k = 1 \textbf{ to } n-1: \quad \textbf{compare neighbours in } 1 \ldots n-k$$

Comparisons are about $n(n-1)/2$, so the growth is $O(n^2)$. The sort is **stable**: equal keys keep their original order. It is fine for teaching and for tiny lists. It is the wrong tool for a large catalogue.

Worked example — do not skip the steps

Board trace of $\{8, 4, 1, 9, 3\}$.

Pass	List at end of pass	Settled
1	4, 1, 8, 3, 9	9
2	1, 4, 3, 8, 9	8, 9
3	1, 3, 4, 8, 9	4, 8, 9
4	1, 3, 4, 8, 9	done

Pass 1 in words: $8 \leftrightarrow 4$, then $8 \leftrightarrow 1$, 8 stays before 9, $9 \leftrightarrow 3$. Speak every swap. A silent ‘it becomes sorted’ earns little.

Silent drill — cover the green model lines

1. After pass 1 of bubble sort, which value is in its final seat?

Model. The maximum. In the example, 9.

2. Stable or not?

Model. Stable.

3. Complexity class?

Model. $O(n^2)$.

4. First swap on $\{8, 4, 1, 9, 3\}$?

Model. 8 and 4, producing $\{4, 8, 1, 9, 3\}$.

5. Why ignore the last cell on pass 2?

Model. It already holds the largest value, so it will not move.

> **TONIGHT** Trace $\{5, 1, 4, 2\}$ on paper. Four lines, one per pass. No calculator.

DAY 15

Selection sort

Unit 2 · Computational Thinking · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Find the next minimum and plant it
- Contrast swap counts with bubble sort
- Say why it is usually not stable

TRAP THAT COSTS THE MARK

Describing selection sort as ‘swap neighbours’. That is bubble sort.

Ten-minute recall — yesterday, closed book

Ask	Accept
Bubble sort: what settles at the end of pass 1?	The largest value
Is bubble sort stable?	Yes

Board notes

On pass i, scan the unsorted suffix, find the smallest item, and swap it into position i. You pay many comparisons and at most one swap per pass.

```
for i = 1 to n-1
    min = i
    for j = i+1 to n
        if A[j] < A[min] then min = j
    swap A[i] with A[min]
```

Same $O(n^2)$ family as bubble sort. Usually **not stable**, because a swap can jump an equal key over another. Prefer it only when writes are expensive and n is small.

Worked example — do not skip the steps

Board on $\{8, 4, 1, 9, 3\}$.

i	Minimum found	List after the swap
1	1	1, 4, 8, 9, 3
2	3	1, 3, 8, 9, 4
3	4	1, 3, 4, 9, 8
4	8	1, 3, 4, 8, 9

Count swaps: four, not a swap on every comparison. That is the contrast with yesterday.

Silent drill — cover the green model lines

1. What is planted at the front on pass 1?

Model. The smallest value in the whole list.

2. At most how many swaps per pass?

Model. One.

3. Complexity?

Model. $O(n^2)$, same family as bubble sort.

4. Stable?

Model. Usually no.

5. Bubble or selection if moving a record is very costly and n is tiny?

Model. Selection, because it swaps at most once per pass.

> **TONIGHT** Selection-sort $\{5, 1, 4, 2\}$ beside yesterday's bubble trace. Circle the swaps.

DAY 16

Linear search and binary search

Unit 2 · Computational Thinking · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Search an unsorted list linearly
- Compute the midpoint
- Refuse binary search when the list is unsorted

TRAP THAT COSTS THE MARK

Sorting ‘in your head’ and still calling it a linear search of unsorted data. If you sorted, say so, then you may search in binary.

Ten-minute recall — yesterday, closed book

Ask	Accept
Selection sort swaps per pass, at most?	One
Bubble complexity?	$O(n^2)$

Board notes

Linear search starts at the left and compares until it hits the key or falls off the end. It works on unsorted data. Worst case $O(n)$.

Binary search needs a sorted array and random access. Keep beginning, end, and midpoint.

$$MID = \left\lfloor \frac{BEG + END}{2} \right\rfloor$$

If the key is smaller than $A[MID]$, throw away the right half. If larger, throw away the left. Repeat. Growth is $O(\log n)$.

Worked example — do not skip the steps

Board. Find 7 in $\{8, 3, 6, 1, 7, 2, 4, 5\}$. Linear: inspect 8, 3, 6, 1, then 7. Five comparisons. Binary is illegal until you sort.

Find 3 in sorted $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Mid index holds 5. Key is smaller, so discard 6–9. Next mid lands on 2 or 3; one more comparison finds 3. Grey cells were never visited. That saving is the whole point.

Silent drill — cover the green model lines

1. Can you binary-search a pile of unmarked scripts?

Model. No. They are not a sorted random-access array.

2. Linear search complexity?

Model. $O(n)$.

3. Binary search complexity, and the condition?

Model. $O(\log n)$, and the array must already be sorted.

4. MID formula?

Model. Floor of (BEG+END)/2.

5. Key smaller than A[MID]. Which half dies?

Model. The right half, including the old mid. END becomes MID−1.

> **TONIGHT** Binary-search 6 in $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Write every MID value.

DAY 17

Correct, efficient, and clear

Unit 2 · Computational Thinking · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Use a trace table
- Pick an algorithm from the situation
- Name a clarity fix

TRAP THAT COSTS THE MARK

Claiming binary search is always faster. It is faster only after the data is sorted, and sorting itself costs work.

Ten-minute recall — yesterday, closed book

Ask	Accept
MID formula?	floor((BEG+END)/2)
Linear search needs sorted data?	No

Board notes

Judge every algorithm three ways.

Test	Question
Correctness	Does it match the specification, including empty input, and does it stop?
Efficiency	How do steps and memory grow with n?
Clarity	Could another student trace it without a phone call?

Decision guide: unsorted search → linear; sorted array search → binary; data must itself be ordered → sort; the same sub-problem returns → remember the answer (the book flags dynamic programming).

A trace table has columns for the line, each variable, the condition, and the output. Use a tiny n.

Worked example — do not skip the steps

Board: sum $1..N$ for $N = 3$. S starts at 0. After $i = 1$, $S = 1$. After $i = 2$, $S = 3$. After $i = 3$, $S = 6$. Then $i = 4$ fails $i \leq 3$ and the loop stops. Closed form $\frac{N(N+1)}{2}$ checks the trace: 6.

Clarity fix: rename `x` to `absentCount` and extract “count absences” from “print the report”.

Silent drill — cover the green model lines

1. Three judgement words.

Model. Correctness, efficiency, clarity.

2. Unsorted bag, find one name. Which search?

Model. Linear.

3. Sorted array of 100000 roll numbers. Which search?

Model. Binary.

4. What does a trace table catch that a shrug does not?

Model. A wrong update, an off-by-one, or a loop that never ends.

5. One clarity repair.

Model. A name that states the unit, such as velocity = distance / time, or one function per job.

> **TONIGHT** Trace ‘count the evens in $\{4, 7, 2, 9\}$’ with columns i, value, count.

DAY 18

Unit 2 checkpoint

Unit 2 · Computational Thinking · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Trace one sort and one search
- Choose the algorithm in one line
- Score 7/10 before Python starts

TRAP THAT COSTS THE MARK

Memorising ‘bubble is slow’ without being able to show one pass.

Ten-minute recall — yesterday, closed book

Ask	Accept
Three tests of an algorithm?	Correct, efficient, clear
Selection: swaps per pass?	At most one

Board notes

Closed book, 40 minutes. Algorithms are earned by traces, not by the name alone. If the sort trace is wrong, Day 19 still happens, but tonight belongs to the trace, not to new Python.

Worked example — do not skip the steps

Mark with the scheme under each item. A correct final list with no passes shown is half a mark at most — this checkpoint wants the passes.

Silent drill — cover the green model lines

1. Define algorithm in one sentence.

Model. A finite, language-free sequence of steps that solves a problem.

2. One pattern and one abstraction in making tea.

Model. Pattern: boil then steep is always the order. Abstraction: ignore cup colour.

3. Bubble-sort pass 1 of $\{3, 1, 2\}$. Show the list.

Model. Compare $3 \leftrightarrow 1 \rightarrow \{1, 3, 2\}$; then $3 \leftrightarrow 2 \rightarrow \{1, 2, 3\}$. 3 has settled.

4. Is bubble sort stable? Is selection sort?

Model. Bubble yes. Selection usually no.

5. Both classroom sorts share which complexity?

Model. $O(n^2)$.

6. Find 2 in unsorted $\{9, 2, 5\}$ and name the search.

Model. Linear. Compare 9, then 2. Two comparisons.

7. Binary search requirement.

Model. A sorted array with random access.

8. Key greater than A[MID]. What changes?

Model. BEG becomes MID+1.

9. What is a trace table for?

Model. A hand execution that checks correctness on a small input.

10. Large sorted file, many lookups. Search choice?

Model. Binary search, $O(\log n)$ per lookup.

> **TONIGHT** Rewrite any lost trace until the swaps are visible. Then stop.

DAY 19

Python values, names, and input

Unit 3 · Programming Fundamentals · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Name a legal variable
- Convert input before arithmetic
- Use an f-string

TRAP THAT COSTS THE MARK

Forgetting int() and then ‘adding’ two ages into a longer string.

Ten-minute recall — yesterday, closed book

Ask	Accept
Binary search needs what first?	Sorted data
Bubble pass 1 settles what?	The maximum

Board notes

A program is a list of instructions. Python is high-level: closer to structured English than to raw bits. A variable is a named box. Names start with a letter or underscore, may contain digits, are case-sensitive, and must not be keywords.

`input()` always returns a **string**. `"3"+"4"` is `"34"`. Wrap with `int()` or `float()` before arithmetic.

```
name = input("Name: ")
age = int(input("Age: "))
print(f"{name} is {age}")
```

Type	Example
int	18
float	1.75
str	"Ali"
bool	True (capital T)

Worked example — do not skip the steps

Board bug. `print(3 + "4")` raises TypeError. Repair: `print(3 + int("4"))` which prints 7. Say why: the quotes make a string, and a string is not a number until you convert it.

Silent drill — cover the green model lines

1. Is `2name` legal?

Model. No. A name cannot start with a digit.

2. What type does input() return?

Model. str, always.

3. Result of int('4') + 3?

Model. 7.

4. Result of '4' + '3'?

Model. The string 43, not 7.

5. Write an f-string printing name and marks.

Model. `print(f"{name} scored {marks}")`.

> **TONIGHT** Type the three-line program and run it. If you have no PC, write the expected screen for age 17.

DAY 20

Operators

Unit 3 · Programming Fundamentals · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Choose // versus /
- Predict a comparison
- Connect and/or/not to Unit 1

TRAP THAT COSTS THE MARK

Using = inside an if condition. That is assignment, not a test.

Ten-minute recall — yesterday, closed book

Ask	Accept
input() returns?	A string
int('3')+int('4')?	7

Board notes

Arithmetic: `+ - * / // % **`. `/` is true division. `//` is floor division. `%` is remainder. `**` is power.

Comparisons `== != > < >= <=` produce `True` or `False`. Logical `and`, `or`, `not` are the same ideas as AND, OR, NOT. Membership: `in`. A single `=` assigns; a double `==` compares.

$$4 == 5 - 1 \Rightarrow \text{True}, \quad 22! = 23 - 1 \Rightarrow \text{False}$$

Worked example — do not skip the steps

Board. `17 // 5` is 3. `17 % 5` is 2. `17 / 5` is 3.4. `2 ** 3` is 8. `55 >= 11 * 5` is True because both sides are 55.

Even test: `n % 2 == 0`. Say it as “remainder 0 when divided by 2”.

Silent drill — cover the green model lines

1. 17 // 5 and 17 % 5?

Model. 3 and 2.

2. What does a single = do?

Model. Assignment. Comparison is ==.

3. not True?

Model. False. Same flip as a NOT gate.

4. 4 == 5 - 1?

Model. True.

5. Expression that is true when code is 'BCA' or 'BCOM'.

Model. `code == 'BCA' or code == 'BCOM'`.

> **TONIGHT** Write five predictions, then run them. Mark any mismatch; that mismatch is a logical or syntax lesson, not a mystery.

DAY 21

Selection

Unit 3 · Programming Fundamentals · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Write if, elif, and else
- Indent the block
- Explain a lone if versus if-else

TRAP THAT COSTS THE MARK

Forgetting the indent so the ‘success’ print always runs.

Ten-minute recall — yesterday, closed book

Ask	Accept
Floor division operator?	//
Remainder operator?	%

Board notes

Selection chooses a block. A colon ends the condition. The next indented lines belong to it. A lone `if` may do nothing when the test is false. `if-else` always runs exactly one of the two blocks. `elif` adds further tests, still only one block runs.

```
age = int(input("Age: "))
if age >= 18:
    print("You can vote")
elif age >= 13:
    print("Teenager")
else:
    print("Too young to vote")
print("Done")
```

`Done` is not indented, so it always prints. That is sequence after selection.

Worked example — do not skip the steps

Board, age 16. First test $16 \geq 18$ is false. Second test $16 \geq 13$ is true, so print Teenager and skip else. Then print Done.

Age 20. First test true, print the vote line, skip elif and else, print Done.

Silent drill — cover the green model lines

1. What character ends the condition line?

Model. A colon.

2. Age 12. Which message, and does Done print?

Model. Too young to vote. Yes, Done prints.

3. Lone if when the test is false?

Model. The indented block is skipped. The program continues underneath.

4. How many blocks run in an if/elif/else chain?

Model. Exactly one.

5. Why convert age with int before comparing to 18?

Model. input() is a string. Comparing a string to 18 is not the numeric test you want.

> **TONIGHT** Write a three-way marks program: 80 and above distinction, 50 and above pass, otherwise retake. Trace marks = 50.

DAY 22

Repetition

Unit 3 · Programming Fundamentals · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Use range stops correctly
- Write a while that ends
- Choose break or continue

TRAP THAT COSTS THE MARK

range(1, 5) when you wanted 1 through 5. The stop is exclusive; use 6.

Ten-minute recall — yesterday, closed book

Ask	Accept
Colon and indent belong to which structure?	if / elif / else, and also loops
Does a lone if always print something?	No

Board notes

`range(a, b)` starts at a and stops **before** b. `range(5)` is 0,1,2,3,4. `range(2, 8)` is 2 through 7. `range(0, 10, 2)` steps by 2.

```
count = 1
while count <= 5:
    print(count)
    count += 1
```

`break` leaves the loop. `continue` skips the rest of this iteration. Nested loops: the inner loop finishes a full cycle for every single outer step, which is why they feel like $O(n^2)$.

Worked example — do not skip the steps

Board: odds from 1 to 7. `for i in range(1, 8, 2): print(i)` prints 1, 3, 5, 7. The stop value 8 is not printed.

Infinite loop to diagnose. `while count <= 5` with `count -= 1` walks away from 5 and never ends. The repair is `+= 1`. This is a logical bug: the program runs, forever, and the answer never appears.

Silent drill — cover the green model lines

1. Values of range(2, 6)?

Model. 2, 3, 4, 5. Not 6.

2. Print 0,2,4,6,8 with one range.

Model. `range(0, 10, 2)` or `range(0, 9, 2)`.

3. break versus continue?

Model. break exits. continue jumps to the next iteration.

4. A while that never adds to the counter.

Model. It does not terminate. Logical error.

5. Why can nested loops be slow?

Model. The inner loop repeats fully for every outer step.

> **TONIGHT** Write a while loop that sums 1 through 5. Trace S. It must finish at 15.

DAY 23

Libraries and the three bug families

Unit 3 · Programming Fundamentals · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Import math three ways
- Name syntax, runtime, and logical errors
- Read a traceback's line number

TRAP THAT COSTS THE MARK

Calling a logical error a syntax error because the answer 'looks wrong'. If it ran, it is not syntax.

Ten-minute recall — yesterday, closed book

Ask	Accept
range(1,4) values?	1, 2, 3
continue does what?	Skips to the next iteration

Board notes

A library is code you did not have to write. Built-in modules this year: `math` (`sqrt`, `pi`, `floor`, `ceil`), `random` (`randint`), `datetime`. Third-party packages install with `pip`.

```
import math
import math as m
from math import sqrt, pi
```

Family	Symptom	Example
Syntax	Will not start	Missing) or colon; bad indent
Runtime	Starts, then crashes	10/0, int('abc')
Logical	Runs, wrong answer	count -= 1 in a while

Worked example — do not skip the steps

Board traceback. `int("abc")` raises ValueError and names the line. You do not 'try random fixes'. You read the type, the file, and the line, then repair that line.

`ImportError` or `ModuleNotFoundError` means the name is wrong or the package was never installed. `math.sqrt(16)` is 4.0, a float.

Silent drill — cover the green model lines

1. Three import styles — name one.

Model. import math; or import math as m; or from math import sqrt.

2. 10/0 is which family?

Model. Runtime. ZeroDivisionError.

3. A missing colon is which family?

Model. Syntax. The program does not start.

4. Wrong average but no crash?

Model. Logical.

5. What does a traceback tell you first?

Model. The exception type and the line number.

TONIGHT Write one example of each bug family on a card. Keep the card for the checkpoint.

DAY 24

Unit 3 checkpoint, on paper

Unit 3 · Programming Fundamentals · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Trace a short program by hand
- Repair one bug and name its family
- Score 7/10

TRAP THAT COSTS THE MARK

Inventing output for a program that would have crashed.

Ten-minute recall — yesterday, closed book

Ask	Accept
Three bug families?	Syntax, runtime, logical
math.sqrt(16)?	4.0

Board notes

No computer required. Hand-tracing is the exam skill. Write the screen, not a paragraph about the screen.

Worked example — do not skip the steps

Mark strictly. A program that would crash must be described as a crash, not as a guessed number.

Silent drill — cover the green model lines

1. Legal or not: `class = 5`?

Model. Illegal. class is a keyword.

2. Screen of print(int('8')+1)?

Model. 9.

3. Screen of print('8'+'1')?

Model. 81.

4. 17 // 5 and 17 % 5.

Model. 3 and 2.

5. Age 18 in the Day 21 program. First message?

Model. You can vote. Then Done.

6. Values printed by for i in range(3): print(i).

Model. 0 then 1 then 2.

7. Family of a missing colon?

Model. Syntax.

8. Family of n % 0 when n is 5? Wait, division by zero via 5/0.

Model. Runtime.

9. Write a condition: pass at 50 or above.

Model. `if marks >= 50:`

10. One line that sums a list named nums using a built-in.

Model. `total = sum(nums)`.

> **TONIGHT** Retype the two programs you traced and run them if a PC exists. The screen must match your paper.

DAY 25

Keys and relationships

Unit 4 · Data and Analysis · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Distinguish data from information
- Place a primary key and a foreign key
- Break M:N with a junction table

TRAP THAT COSTS THE MARK

Putting the foreign key on the ‘one’ side and wondering why many loans will not fit.

Ten-minute recall — yesterday, closed book

Ask	Accept
Checkpoint target?	7/10
range stop is inclusive?	No, it stops before the end value

Board notes

Data are raw facts. Information is data after processing. A database is an organised store. A DBMS (Access, MySQL, SQL Server) reduces redundancy and enforces rules.

A **primary key** uniquely identifies a row. A **foreign key** in another table stores that key so the rows belong together. Cardinality:

Link	How it is stored
1:1	The key of A stored uniquely in B
1:M	The foreign key sits on the many side
M:N	Illegal as a single link; add a junction table

Referential integrity: a foreign key must match an existing primary key, or be empty if the design allows it. Cascade update rewrites child keys. Cascade delete removes child rows — dangerous if you still need history.

Worked example — do not skip the steps

Board. One student, many loans. STUDENT(StudentID PK, Name). LOAN(LoanID PK, StudentID FK, BookID FK, DueDate). The 1 is on STUDENT. The M is on LOAN. You cannot store a repeating group of book ids inside STUDENT and stay in a clean relational design.

Students and classes are M:N. Junction: ENROLMENT(StudentID, ClassID), both foreign keys.

Silent drill — cover the green model lines

1. Marks 70, 80, 90. One sentence of information.

Model. The mean is 80, so the class met a pass line of 80. The sentence is the information; the three numbers are data.

2. Where does the foreign key sit in 1:M?

Model. On the many side.

3. Why a third table for M:N?

Model. Each pair is stored once. A single table cannot hold a repeating group of keys cleanly.

4. Cascade delete of a member who still has loans?

Model. The loan rows disappear too. Wrong if history matters.

5. Candidate key versus primary key?

Model. Any minimal unique field is a candidate. You promote one of them to primary; the others are alternate keys.

TONIGHT Draw STUDENT and LOAN with keys labelled PK and FK. One minute.

DAY 26

The library ER and Access objects

Unit 4 · Data and Analysis · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Draw Book, Member, Borrow
- Name four Access objects
- Choose Design View for a real table

TRAP THAT COSTS THE MARK

Storing the member’s name again inside every loan row. That copy drifts; store the name once and point at it with the key.

Ten-minute recall — yesterday, closed book

Ask	Accept
M:N needs what?	A junction table
FK sits on which side of 1:M?	The many side

Board notes

Entities are things we store. Attributes are properties. Relationships are verbs. The library case cannot link Member directly to Book as M:N, so BorrowRecord is the junction.

Book(BookID, Title, Author), Member(MemberID, Name), Borrow(BorrowID, MemberID, BookID, dates)

Access objects: **tables** store rows, **forms** are the entry screens, **queries** ask questions, **reports** print what a principal reads. Datasheet View is quick and weak. Design View is where you name fields, pick types, and set the primary key (AutoNumber for StudentID, Short Text for names, Number for age).

Worked example — do not skip the steps

Board query in words. “BCA students sorted by name.” That is a query, not a new table: filter Department = "BCA", sort Name, show only the columns you need. A report can sit on that query. A form can sit on the Student table so clerks do not type into the grid.

Silent drill — cover the green model lines

1. Three library entities.

Model. Book, Member, BorrowRecord (or Loan).

2. Which fields in Borrow are foreign keys?

Model. MemberID and BookID.

3. Four Access objects.

Model. Tables, forms, queries, reports.

4. Design View or Datasheet View to set a primary key properly?

Model. Design View.

5. A filter on Department is which object?

Model. A query.

> **TONIGHT** Sketch the three boxes and the 1 and M labels. Underline every primary key.

DAY 27

IoT, AI, sources, and the human cost

Unit 5 · Applications and Impacts · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Draw the smart loop
- Separate primary, secondary, and tertiary
- Name one assistive tool

TRAP THAT COSTS THE MARK

Calling a chatbot ‘the Internet of Things’. A chatbot is AI. IoT needs a physical device.

Ten-minute recall — yesterday, closed book

Ask	Accept
Access objects?	Table, form, query, report
Junction entity in the library?	BorrowRecord

Board notes

Three technologies travel together. **IoT** devices sense and act. **Analytics** cleans and finds patterns. **AI** decides. A hot room: temperature sensor, a trend check, then the AC actuator.

sensor → connect → analyse → decide → actuator

Sources: **primary** is first-hand (your interview, an official register). **Secondary** is already processed (a textbook, a statistics report). **Tertiary** summarises summaries (many encyclopaedia pages). Wikipedia is not a primary source.

Assistive technology is part of the subject: screen readers such as NVDA, captions, AAC boards, switch controls. Every gain (records, maps, LMS) has a cost (privacy, fatigue, people left offline).

Worked example — do not skip the steps

Board: smart irrigation. Soil-moisture sensor, LoRa or mobile data, a cleaned reading, a rule or a model, a valve actuator. Name the sensor and the actuator in the answer or the loop is incomplete.

Source drill: “I surveyed my own class” is primary. “I quoted the Pakistan Bureau of Statistics” is secondary. “I used an unsigned blog that summarises both” is weak tertiary.

Silent drill — cover the green model lines

1. One sentence each: IoT, analytics, AI.

Model. IoT senses and acts in the world. Analytics finds patterns. AI decides or predicts.

2. Sensor and actuator in an automatic door.

Model. Motion sensor; motor that opens the door.

3. Is a Wikipedia overview primary?

Model. No. It is tertiary, sometimes secondary. It summarises other work.

4. One positive and one negative of digital school records.

Model. Faster lookup for a doctor or teacher; privacy harm if the file leaks.

5. Name one screen reader.

Model. NVDA or VoiceOver.

> **TONIGHT** Draw the five-box loop and hang a farm example on it. Six lines maximum.

DAY 28

A research day you can finish

Unit 6 · Digital Literacy · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Write one research question
- Separate qualitative from quantitative
- Search with operators and cite

TRAP THAT COSTS THE MARK

A question so wide (‘all students in Pakistan’) that no method can answer it this week.

Ten-minute recall — yesterday, closed book

Ask	Accept
IoT actuator does what?	Turns a decision into a physical action
Primary source in one line?	First-hand data collected for this question

Board notes

Digital literacy is judgement, not device ownership. Five verbs: find, collect, judge, present, cite.

	Qualitative	Quantitative
Form	Reasons, opinions	Counts, hours, marks
Use	Why	How many, how much

Primary data is what you collect for this question. Secondary data is what you reuse. Methods: interview, survey, observation, prototype, simulation.

Search operators: quotes for a phrase, `filetype:pdf`, `site:.edu` or `site:.gov.pk`. Copying without credit is plagiarism. Asking permission is part of the method.

Worked example — do not skip the steps

Board question. “How does daily phone use affect study habits of Class XI in this college?” Not the topic “phones”. Sample: one section, not all of Pakistan. Tools: a Form, a Sheet, one chart, a one-page report. Ethics: no names you do not need.

Best query of the set: `"internet usage in Pakistan" filetype:pdf`. The vague query `best internet social media` is how rumours enter the bibliography.

Silent drill — cover the green model lines

1. Topic versus research question. Repair ‘social media’.

Model. How many hours do Class XI students in this college spend on social media on a school night, and what do they say it does to revision?

2. Survey of your own class: primary or secondary?

Model. Primary.

3. A government PDF you did not collect: which type?

Model. Secondary.

4. Which operator keeps words together?

Model. Quotation marks around the phrase.

5. The offence of pasting a paragraph with no credit.

Model. Plagiarism.

> **TONIGHT** Write the question, four quantitative items, and two qualitative items. Stop. That is a complete crash artefact outline.

DAY 29

Full mock — 90 minutes, 50 marks

Mock paper · Class XI · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Attempt every question
- Show traces and tables, not only final words
- Write the time on the script

TRAP THAT COSTS THE MARK

Adding notes in the margin after the time is called. Day 30 will not count those lines.

Sit this as an exam. 90 minutes. No notes. No phone.

SECTION A · 10 × 1 = 10 · ONE WORD OR ONE LINE

1. Boolean value of $1 + 1$.
2. Gate that outputs 1 only when inputs differ.
3. Number of rows in a truth table with 4 inputs.
4. PDU at the network layer.
5. TCP/IP layer that contains Wi-Fi.
6. Search that requires sorted data.
7. Python type returned by `input()`.
8. Operator for remainder.
9. Access object used to filter rows.
10. Copying text with no credit is called what?

SECTION B · 8 × 3 = 24 · SHOW WORKING

11. Draw the truth table of $Y = \overline{A} + B$.
12. Simplify $Y = \overline{A}B + AB$ and name the method (algebra or K-map).
13. Give one reason the Al-Noor library used Waterfall and one reason the mobile app used Agile.
14. List OSI layers 4, 3, and 2 with the address or number each one adds (port, IP, or MAC).
15. Bubble-sort pass 1 only of $\{6, 2, 5, 1\}$. Show the list after the pass and name the settled value.
16. Write a Python `if/elif/else` that prints Pass at 50 or above and Fail otherwise. Assume `marks` is already an int.
17. A student takes many classes and a class has many students. Name the junction table and its two foreign keys.
18. Write one research question about phone use that a single college class could actually answer. Label one quantitative item you would collect.

SECTION C · 2 × 8 = 16 · STRUCTURED PARAGRAPHS OR TRACES

19. Explain encapsulation as a web request moves down the TCP/IP stack. Name the data unit at each of the four layers. (8)

20. A program is meant to sum 1 through 5 but the screen never returns. The loop is `while n <= 5: total += n` and `n` starts at 1. Name the bug family, repair the code, and show a two-column trace of `n` and `total` after the repair. (8)

> **STOP** Do not open Day 30 until the paper is finished and the time is written on the top.

> **TONIGHT** Sleep. Bring the script to Day 30 unmarked.

DAY 30

Mark the mock and build the 48-hour card

Mock conference · Class XI · 120-minute coached session

10 min	25 min	25 min	40 min	20 min
Recall, cold	Teach from the board	Work the example together	Silent drill, then mark	Homework set and traps

LEAVE ABLE TO

- Mark to the scheme, not to kindness
- Sort lost marks into three repair piles
- Leave with a one-page card

TRAP THAT COSTS THE MARK

Re-marking generously. A blank trace is not ‘almost’.

Marking the mock — coach aloud, student in a different colour

Section A — one mark, no method mark

1. **1** (OR, not 2).
2. XOR.
3. **16** because $2^4 = 16$.
4. Packet.
5. Network access.
6. Binary search.
7. `str` / string.
8. `%`.
9. Query.
10. Plagiarism.

Section B — three marks, award points as written

11. Four rows present (1). $\overline{A}$ column correct (1). Y is 0 only on row 1,0 (1).
12. $Y = B$ (2). Method named: factor $B(\overline{A} + A)$ or a 2-cell group on the map (1).
13. Library: known, signed, frozen requirements (1). App: unclear needs or many stakeholders or early demo (1). Names of the two projects (1).
14. Transport / port (1). Network / IP (1). Data link / MAC (1).
15. Swaps shown through to $\{2, 5, 1, 6\}$ (2). Settled value named as 6 (1).
16. Colon and indent (1). `>= 50` (1). Both messages, else included (1).
17. Junction named, e.g. Enrolment (1). StudentID FK (1). ClassID FK (1).
18. A question with a population and a measurable idea (2). One numeric item such as hours per night (1).

Section C — eight marks

19. Application data / HTTP (2). Transport segment / TCP (2). Internet packet / IP (2). Network access frame then bits (2). “OSI instead of TCP/IP” with correct jobs may earn the job marks but not the layer-name marks.

20. Logical error, loop does not change `n` (2). Repair `n += 1` inside the loop (2). Trace reaches total 15 (2). Stops when `n` becomes 6 (2).

48-HOUR CARD — SYSTEMS

$1 + 1 = 1$. NAND/NOR universal. XOR differs. SOP from 1-rows. Gray code 00, 01, 11, 10. PDU: data, segment, packet, frame, bits. TCP/IP folds 7–5 and 2–1.

48-HOUR CARD — ALGORITHMS & PYTHON

Bubble settles the max, stable, $O(n^2)$. Selection one swap a pass. Binary needs sorted data, MID = floor((BEG+END)/2). input() is str. // and %. range stops early. Bugs: syntax, runtime, logical.

48-HOUR CARD — DATA & SOCIETY

FK on the many side. M:N needs a junction. Access: table, form, query, report. Sensor → decide → actuator. Primary = you collected it. Quotes, filetype, site:. Plagiarism is uncredited copying.

Target reading of the score: 40–50 safe, 30–39 repair the largest pile only, under 30 repeat Days 12, 18, and 24 before any new past paper.

TONIGHT Copy the 48-hour card by hand onto one sheet. That sheet is the only note you carry into the last evening.